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Three Benchmarks Addressing Open Challenges in Nonlinear System Identification ⋆

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Abstract: Nonlinear system identification is a fast evolving field of research with contributions from different communities. It is not always straightforward to compare different models and identification approaches. Therefore it is of high importance to offer well chosen benchmarks to the nonlinear system identification community. The interaction generated by such a benchmark can push the state of the art in nonlinear system identification forward.
This paper discusses some challenges that are present nowadays in the nonlinear system identification community. Three benchmarks are presented, a hysteretic benchmark, a Wiener-Hammerstein benchmark, and a cascaded tanks benchmark, each addressing one or more of these challenges.

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Keywords: Nonlinear System Identification, Benchmark Systems, Hysteresis, Wiener-Hammerstein, Process Noise, Cascaded Tanks, Short Data Record

1. INTRODUCTION

Nonlinear system identification is a fast evolving field of research with contributions from different communities, such as the mechanical engineering, systems and control and machine learning communities. Many identification methods and algorithms have been developed over the last years, for a wide variety of model structures. Block-oriented structures (Giri and Bai, 2010; Schoukens et al., 2015), nonlinear state-space representations (Paduart et al., 2010; Schön et al., 2011), NARX and NARMAX models (Billings, 2013) and piecewise linear models (Mattsson et al., 2016) are some examples of different nonlinear system representations.

It is not always straightforward to compare different models and identification approaches due to the difference between the model structures, the many possible trade-offs (e.g. model complexity, model flexibility, computational load, accuracy and interpretability of the model), the differences between the assumptions that are made and the different backgrounds of the communities involved in nonlinear system identification. Carefully selected and well described benchmarks are of great importance for the validation and comparison of newly developed nonlinear identification algorithms. A benchmark using a real-life system can also be used to validate the robustness of an identification algorithm with respect to non-idealities that are typically encountered in a practical measurement setup and a real-life system (e.g. system does not belong exactly to the model set, slightly violated noise assumptions, non-idealities in the measurements). A well constructed benchmark allows a researcher to more easily understand the strengths and weaknesses of each identification algorithm and the differences between the proposed approaches. Moreover, a well chosen benchmark can increase the interaction and collaboration between the different identification communities by working side by side on the same identification challenge.

Only a limited number nonlinear system identification benchmark datasets are publicly available (see for instance (Wigren and Schoukens, 2013; Schoukens et al., 2009)). Some more interesting setups are collected and described in (Kroll and Schulte, 2014), where the focus lies, depending on the considered benchmark, on identification and/or control.

We believe that it is of crucial importance not to see a benchmark in this setting as a competition, but rather as an opportunity to illustrate the capabilities and limitations of an identification method. Such an interaction can drive the nonlinear system identification field forward, by identifying some common shortcomings in the state of the art, or by combining two or more state-of-the-art approaches to obtain an even more advanced identification algorithm.

This paper introduces three new nonlinear system identification benchmarks to a wide audience. It also serves as an introduction paper on the IFAC 2017 open invited track on Nonlinear System Identification Benchmarks. A discussion on the type of identification approaches that can be applied on the presented benchmarks and datasets is beyond

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the scope of this paper. The remainder of this paper first discusses some challenges that are present nowadays in the nonlinear system identification community (Section 2). Three benchmarks, a hysteretic benchmark with a dynamic nonlinearity (Noël and Schoukens, 2016), a Wiener-Hammerstein benchmark with process noise (Schoukens and Noël, 2016), and a cascaded tanks benchmark combining soft and hard nonlinearities (Schoukens et al., 2016), addressing one or more of these challenges are introduced next in Sections 3, 4 and 5.

2. SOME CHALLENGES IN NONLINEAR SYSTEM IDENTIFICATION

There are quite some challenges to tackle in the nonlinear system identification community. This paper highlights three of them: dynamic nonlinearities, process noise and short data records.

This paper does not offer any solution to these benchmarks, many possible approaches can be applied, and obtain satisfying results. The benchmarks presented in this paper allows us to better understand the advantages and drawbacks of new state-of-the-art nonlinear identification approaches.

2.1 Dynamic Nonlinearities

Dynamic nonlinearities such as hysteretic nonlinearities appear in many engineering disciplines ranging from solid mechanics (Morrison et al., 2001), electromagnetism (Bertotti, 1985) and aerodynamics (Mueller, 1985). A hysteresis nonlinearity is governed by internal states, not accessible from the measured input or output. The (black-box) identification of hysteretic nonlinearities is studied recently using, for example, augmented Hammerstein structures (Wang et al., 2012; Yong et al., 2015), NARX (Worden and Barthorpe, 2012), Neural Network based structures (Xie et al., 2013) and nonlinear state-space structures (Noël et al., 2016).

The challenge of identifying a dynamic system with a hysteretic nonlinearity in feedback is addressed by the benchmark presented in Section 3.

2.2 Process Noise

Many nonlinear identification methods are limited to an additive (colored) noise source located at the output of the system, or are restricted to a NARX or NARMAX type of noise model. Such a noise framework is a simplified representation of reality which can lead to biased estimates, e.g. due to process noise passing through a nonlinear subsystem (Hagenblad et al., 2008). A more realistic noise framework can be obtained by introducing multiple noise sources, or by placing the noise source at a different location in the considered system structure. Multiple noise sources are considered in a limited number of contributions: e.g. (Hagenblad et al., 2008; Wills et al., 2013; Lindsten et al., 2013; Wahlberg et al., 2014). This more realistic noise framework comes often at the cost of a more complex identification algorithm.

The Wiener-Hammerstein benchmark with process noise presented in Section 4 challenges the research community to deal with a dominant process noise source during the identification of a Wiener-Hammerstein system.

2.3 Short Data Records

Classical (black-box) nonlinear system identification methods often require a relatively high amount of data to obtain a high quality model which demonstrates good generalization capabilities on a validation data record. The number of data points needed can be reduced by introducing prior knowledge, or physical insight into the model structure (e.g. through regularization (Risuleo et al., 2015) or using a structured (grey-box) models).

The challenge of identifying a nonlinear system using short data records is addressed by the cascaded tanks benchmark combining soft and hard nonlinearities in Section 5.

3. HYSTERETIC BENCHMARK WITH A DYNAMIC NONLINEARITY

A more in-depth description of the hysteretic benchmark with a dynamic nonlinearity is provided in (Noël and Schoukens, 2016).

3.1 System

The Bouc-Wen model (Bouc, 1967; Wen, 1976) has been intensively used to model hysteretic effects in mechanical engineering, especially in the case of random vibrations. An extensive literature review about Bouc-Wen modeling can be found in (Ismail et al., 2009; Ikhouane and Rodellar, 2007).

The vibrations of a single-degree-of-freedom Bouc-Wen system, i.e. a Bouc-Wen oscillator with a single mass, are given by (Wen, 1976)

$$m_L\,\ddot{y}(t) + r(y,\dot{y}) + z(y,\dot{y}) = u(t), \tag{1}$$

where m_L is the mass constant, y the displacement, u the external force, and where an over-dot indicates a derivative with respect to the time t. The restoring force in the system is composed of a static nonlinear term $r(y,\dot{y})$, which only depends on the instantaneous values of the displacement $y(t)$ and velocity $\dot{y}(t)$, and of a dynamic nonlinear term $z(y,\dot{y})$, which represents the hysteretic memory of the system. Here we assume that the static restoring force contribution is linear:

$$r(y,\dot{y}) = k_L\,y + c_L\,\dot{y}, \tag{2}$$

where k_L and c_L are the linear stiffness and viscous damping coefficients, respectively. The hysteretic force $z(y,\dot{y})$ obeys the first-order differential equation

$$\dot{z}(y,\dot{y}) = \alpha\,\dot{y} - \beta\left(\gamma\,|\dot{y}|\,|z|^{\nu-1}\,z + \delta\,\dot{y}\,|z|^{\nu}\right), \tag{3}$$

where the five Bouc-Wen parameters α, β, γ, δ and ν are used to tune the shape and the smoothness of the system hysteresis loop. Table 1 lists the values of the physical parameters selected in this study. The system parameters m_L, c_L and k_L given in Table 1 result in a linear modal natural frequency $\omega_0 = 35.59$Hz and a damping ratio $\zeta = 1.12\%$. The Bouc-Wen parameter values in Table 1 were selected to lead to appreciable nonlinear manifestations under reasonable forcing levels. More specifically, in the two test data sets described below, the natural frequency

Table 1. Physical parameters of the Bouc-Wen system.

Parameter	m_L	c_L	k_L	α	β	γ	δ	ν
Value (SI unit)	2	10	$5\,10^4$	$5\,10^4$	$1\,10^3$	0.8	-1.1	1

and damping ratio of the system are altered in nonlinear regime of vibration by approximately 10 % (hardening stiffness) and 500 % (hysteretic damping), respectively. Fig. 1 (a) illustrates the existence of a non-degenerate steady-state loop in the system input-output plane for quasi-static forcing conditions. In comparison, by setting the β parameter to 0, a linear steady-state behaviour is retrieved in Fig. 1 (b). In these two figures the system is excited by a sine wave with a frequency of 1 Hz and an amplitude of 120 N.

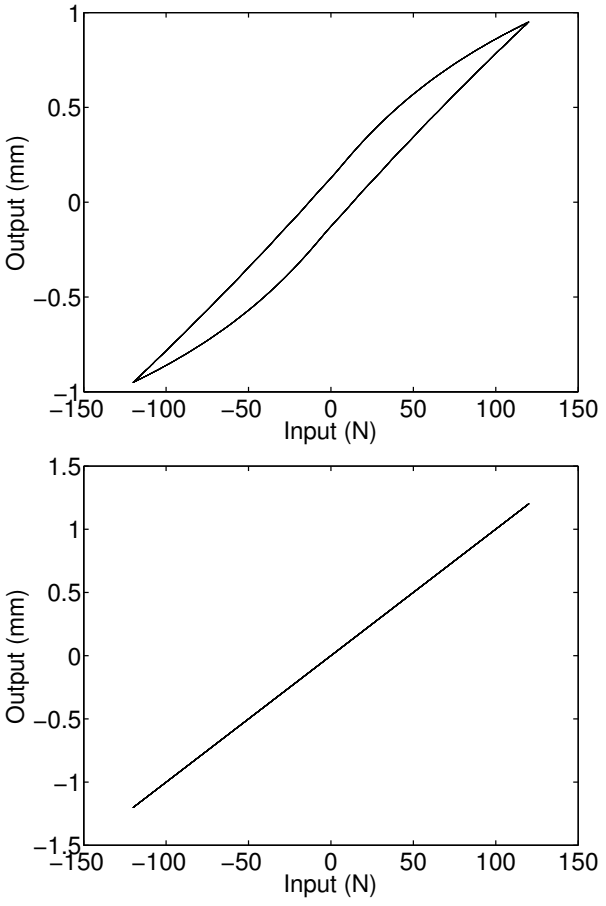


Fig. 1. Hysteresis loop in the system input-output plane for quasi-static forcing conditions. (a) Non-degenerate loop obtained for the parameters in Table 1; (b) linear behavior retrieved when setting the β parameter to 0.

3.2 Identification Challenges

We believe that there are 4 major nonlinear system identification challenges to be associated with the Bouc-Wen benchmark:

(1) it possesses a dynamic nonlinearity, *i.e.* a nonlinearity with memory;
(2) the internal variable $z(t)$ governing the nonlinearity is not accessible by the user;
(3) the nonlinear functional form in eq. (3) is nonlinear in the parameter ν;
(4) the nonlinear functional form in eq. (3) does not admit a finite Taylor series expansion because of the presence of absolute values.

3.3 Available Data

The Bouc-Wen benchmark is available as a Matlab simulation package, together with a minimal working example through the benchmark website (`http://homepages.vub.ac.be/~mschouke/benchmarkBoucWen.html`). A detailed description of the simulation setup and some guidelines for the users are given in (Noël and Schoukens, 2016). Benchmark users have the freedom to apply as many self-designed signals as desired.

Two fixed test datasets are provided through the benchmark website: a random phase multisine and a sine-sweep signal. The test datasets are noiseless, sampled at 750 Hz. The random phase multisine dataset contains one steady-state period of 8192 samples. The excited band ranges from 5 to 150 Hz, and the input has an RMS-value of 50 N. The sine-sweep dataset started with initial conditions equal to zero, it is not in steady state. In this case, the amplitude of the input is 40 N, and the frequency band from 20 to 50 Hz is covered at a sweep rate of 10 Hz/min. These test sets function as a target for the obtained model. The test data should not be used for any purpose during the estimation.

4. WIENER-HAMMERSTEIN BENCHMARK WITH PROCESS NOISE

A detailed description of the Wiener-Hammerstein benchmark with process noise, the benchmarked system, the measurement setup and the input-output data is provided in (Schoukens and Noël, 2016). A short recapitulation of the most important parts is given in this section.

4.1 System

The Wiener-Hammerstein structure is a well known block-oriented system. It contains a static nonlinearity that is sandwiched in between two LTI blocks (Figure 2). The fact that the static nonlinearity is not directly accessible from the input and output results in a problem that is harder to identify. The system is quite similar to the Wiener-Hammerstein system that is studied in an earlier benchmark (Schoukens et al., 2009; Vandersteen, 1997), the main difference is the presence of the process noise.

The first filter $R(s)$ is a third order lowpass filter. The second LTI subsystem $S(s)$ is designed as an inverse Chebyshev filter with a stopband attenuation of 40dB and a cutoff frequency of 5 kHz. The second LTI subsystem has a transmission zero within the excited frequency range. This makes the inversion of $S(s)$ difficult. The static nonlinearity $f(x)$ is a double sided saturation nonlinearity realized with a diode-resistor network.

The additive process noise $e_x(t)$ is a filtered white Gaussian noise sequence. The filtered noise is generated starting from a discrete-time 3rd order lowpass Butterworth filter

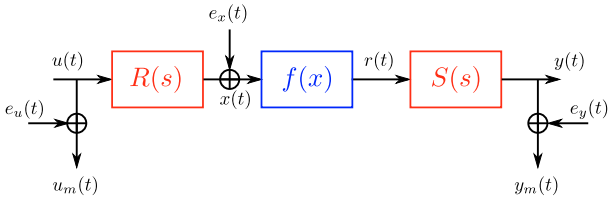


Fig. 2. A Wiener-Hammerstein system with process noise. The LTI blocks at the input and the output are depicted by $R(s)$ and $S(s)$ respectively. $f(x)$ denotes the static nonlinearity. The process noise $e_x(t)$ enters the system before the static nonlinearity. Two smaller (neglectable) noise sources $e_u(t)$ and $e_y(t)$ are present in the measurement channels. $u_m(t)$ and $y_m(t)$ are the measured input and output signals.

followed by a zero-order hold reconstruction and an analog low-pass reconstruction filter with a cut-off frequency of 20kHz. The noise sources $e_u(t)$ and $e_y(t)$ account for the measurement noise, they can be considered to be white Gaussian noise sources. The dominant noise source is $e_x(t)$, the measurement noise is very small.

4.2 Identification Challenges

Three major nonlinear system identification challenges are associated with the Wiener-Hammerstein benchmark with process noise:

(1) the process noise that is present in the system,
(2) the static nonlinearity which is not directly accessible from neither the measured input or output,
(3) the output dynamics are difficult to invert due to the presence of a transmission zero.

4.3 Available Data

An estimation and validation dataset and a test dataset are available for download on `http://homepages.vub.ac.be/mschouke/benchmarkWienerHammerstein.html`. The estimation and validation dataset is acquired based upon the request of the researchers that previously worked on this benchmark. The inputs used are all sampled at 718125 Hz, but vary in type and length:

- Multisine input (10 realization, 2 periods, 1 amplitude, 4096 points per period)
- Swept sine input (100 periods, 64161 points per period)
- Zero input (4096 points, 2 realizations)
- Sine input (at 20 different frequencies, 5 different amplitudes, sequence length: 65536 points)
- Combined multisine, swept sine and zero input (5 realizations, 57346 points per realization)
- Triangular shaped multisine input (32 realizations, 32 periods, 2048 points per period)
- Triangular shaped multisine input (50 realizations, 4 periods, 16384 points per period)

The test dataset is a noiseless dataset which should not be used during the model estimation and validation procedure, it comprises two types of signals:

- Noiseless multisine input (1 period, 16384 points)
- Noiseless swept sine input (1 period, 16384 points)

5. CASCADED TANKS BENCHMARK COMBINING SOFT AND HARD NONLINEARITIES

An in-depth description of the cascaded tanks benchmark is provided in (Schoukens et al., 2016). The most important parts are recapitulated in this section.

5.1 System

The cascaded tanks system is a fluid level control system consisting of two cascaded tanks with free outlets, the upper tank is fed by a pump. The input signal controls the water pump that pumps the water from a reservoir into the upper water tank. The water of the upper water tank flows through a small opening into the lower water tank, and finally through a small opening from the lower water tank back into the reservoir. This process is shown in Figure 3.

The relation between (1) the water flowing from the upper tank to the lower tank and (2) the water flowing from the lower tank into the reservoir are weakly nonlinear functions. However, when the amplitude of the input signal is too large, an overflow can happen in the upper tank, and, with a delay, also in the lower tank. When the upper tank overflows, part of the water goes into the lower tank, the rest flows directly into the reservoir. This effect is partly stochastic, hence it acts as an input-dependent process noise source. The overflow saturation nonlinear behavior of the lower tank is clearly visible in the time domain representation of the output signals (see Figure 4). A video of such an overflow situation can be found on the benchmark website (`http://homepages.vub.ac.be/mschouke/benchmarkCascadedTanks.html`).

Without considering the overflow effect, the following approximate input-output model can be constructed based on Bernoulli's principle and conservation of mass:

$$\dot{x}_1(t) = -k_1\sqrt{x_1(t)} + k_4 u(t) + w_1(t), \tag{4}$$

$$\dot{x}_2(t) = k_2\sqrt{x_1(t)} - k_3\sqrt{x_2(t)} + w_2(t), \tag{5}$$

$$y(t) = x_2(t) + e(t), \tag{6}$$

where $u(t)$ is the input signal, $x_1(t)$ and $x_2(t)$ are the states of the system, $w_1(t)$, $w_2(t)$ and $e(t)$ are additive noise sources, and k_1, k_2, k_3 and k_4 are constants depending on the system properties.

5.2 Identification Challenges

We anticipate the cascaded tanks benchmark to be associated with 4 major nonlinear system identification challenges:

(1) the hard saturation nonlinearity combined with the weakly nonlinear behavior of the system in normal operation,
(2) the overflow from the upper to the lower tank, this effect also introduces input-dependent process noise,
(3) the relatively short estimation data record,
(4) the unknown initial values of the states.

5.3 Available Data

The input signals are multisine signals which are 1024 points long, and excite the frequency range from 0

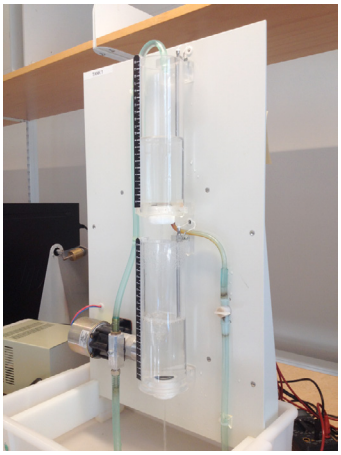


Fig. 3. The cascaded tanks system: the water is pumped from a reservoir in the upper tank, flows to the lower tank and finally flows back into the reservoir. The input is the pump voltage, the output is the water level of the lower tank.

to 0.0144Hz, both for the estimation case and test case. The input and output signals are shown in Figure 4. The lowest frequencies have a higher amplitude then the higher frequencies. The sample period T_s is equal to 4s. The input signals are zeroth-order hold input signals. The cascaded tanks data is available for download on `http://homepages.vub.ac.be/mschouke/benchmarkCascadedTanks.html`.

The process is controlled from a Matlab interface to the A/D and D/A converters attached to the water level sensor and the pump actuator. The water level is measured using capacitive water level sensors, the measured output signals have a signal-to-noise ratio that is close to 40 dB. The water level sensors are not calibrated and can introduce an extra source of nonlinear behavior.

Note that the system was not in steady state during the measurements. The system states have an unknown initial value at the start of the measurements. This unknown state is the same for both the estimation and the test data record.

6. FIGURE OF MERIT

We expect that the RMS of the residuals for all test datasets are reported in the benchmark results to allow for a clear comparison between different methods:

$$\boldsymbol{e_{RMSt}} = \sqrt{1/N_t \sum_{t=1}^{N_t} (y_{mod}(t) - y_t(t))^2}, \qquad (7)$$

where y_{mod} is the modeled output, y_t is the output provided in the test dataset, N_t is the total number of points in y_t.

Users of the benchmark datasets are free to specify, on top of this, other figures of merit that correspond better to their modeling goals.

7. CONCLUSION

This paper introduced three distinct benchmarks, each addressing one main challenge in nonlinear system identification: dynamic nonlinearities, process noise and short data records. A more detailed description of the benchmarks is available.

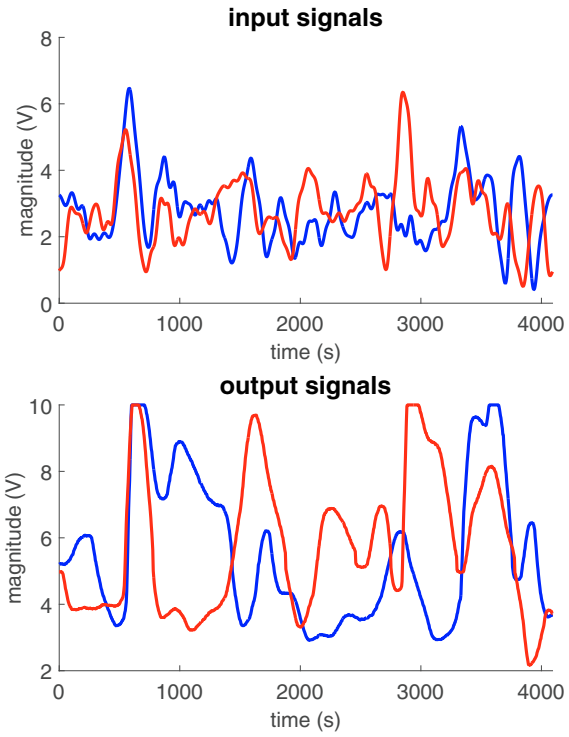


Fig. 4. Input and output signals of the estimation (blue) and test (red) data records in the time domain.

The benchmarks are not meant as a competition, but rather as an opportunity to illustrate the capabilities and limitations of an identification method. With these benchmarks we hope to foster an interaction between the systems and control, machine learning and mechanical communities working on nonlinear data-driven modeling problems. Such an interaction can drive the nonlinear system identification field forward, by identifying some common shortcomings in the state of the art, or by combining two or more state-of-the-art approaches to obtain an even more advanced identification algorithm.

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